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Roll No.

337553(37)

BE (5th Semester)
Examination, Nov.-Dec., 2015

(New Scheme)

Dynamics of Machines

Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

Note : (*i*) Parts (*a*) and (*b*) of each question are compulsory and attempt any **one** from Parts (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

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1. (*a*) Explain the term 'height of a governor'. 2

(*b*) What is meant by effort of a governor? Find the expression for the same in a Porter governor. 4

(*c*) The upper arms of a Porter governor are pivoted on the axis of rotation, their length being 30 cm. The lower arms are pivoted on the sleeve of a distance of 3 cm from the axis, their length being 27 cm

Mass of each ball is 6 kg and the sleeve mass is 50 kg. Determine the equilibrium speed for a radius of rotation of 17 cm and also the effort and power for 1% change of speed. 10

(d) A Hartnell governor having a central sleeve spring and two right-angled bell crank levers moves between 290 r.p.m. and 310 r.p.m. for a sleeve lift of 15 mm. The sleeve arms and ball arms are 80 mm and 120 mm respectively. The levers are pivoted at 120 mm from the governor axis and the mass of each ball is 2·5 kg. The ball arms are parallel to the governor axis at the lowest equilibrium speed. Determine—

(i) loads on the spring at the lowest and highest equilibrium speeds;

(ii) stiffness of the spring. 10

2. (a) Explain, why balancing of machines is necessary. 2

(b) Explain the method of balancing a number of masses revolving in different planes. 4

(c) A shaft carries four masses A, B, C and D in order along a shaft. The masses at B and C are 18 kg and 12·5 kg respectively and each has an eccentricity of 6 cm. The masses at A and D have an eccentricity of 8 cm. The angle between masses at B and C is

100° and that between the masses at B and A is 190°, both the angles are measured in same sense. The axial distance between the planes A and B is 10 cm and between B and C is 20 cm. If the shaft is in complete dynamic balance, determine—

(i) the weights of masses A and D;

(ii) the distance between planes C and D;

(iii) angular position of mass at D. 10

(d) A single cylinder engine runs at 250 r.p.m. and has a stroke of 180 mm. The reciprocating parts have a mass of 120 kg and revolving parts are equivalent to a mass of 70 kg at a radius of 90 mm. A mass is placed outside the crank at a radius of 150 mm to balance the whole of the revolving mass and two-third of the reciprocating mass. Determine the magnitude of balancing mass and the resultant residual unbalance force when the crank has turned 30° from IDC. Neglect the obliquity of connecting rod. 10

3. (a) Define a gyroscope. 2

(b) Explain gyroscopic effect during pitching and rolling of naval ships. 4

(c) An aeroplane flying at 240 km/h turns towards right and completes a quarter circle of 60 m radius.

The mass of rotary engine and the propeller of the plane amounts to 450 kg with a radius of gyration of 320 mm. The engine speed is 2000 r.p.m. clockwise when viewed from the rear. Determine the gyroscopic couple on the aircraft and state its effect. In what way is the effect changed when—

(i) the aeroplane turns towards left;

(ii) the engine rotates clockwise when viewed from the front (nose) and aeroplane turns towards (*1*) left and (*2*) right? 10

(d) A racing car of mass 2500 kg has a wheel base of 2 m and track width of 105 cm. The centre of gravity lies mid-way between the front and rear axles and is 0·4 m above the ground. The engine rotating parts are equivalent of a flywheel of MI of 50 kg-m^2 rotating at 6000 r.p.m. in clockwise direction when viewed from the front. If the car speeding at 50 km / hr rounds a curve of 15 m radius, determine the reaction between the four wheels and the ground. Consider the gyroscopic effect of the flywheel, the dead weight of the car and the centrifugal effect. Neglect the effect due to rotating wheels. 10

4. *(a)* Define damping coefficient. 2

(b) Distinguish between longitudinal, transverse and torsional vibration. State the nature of stress induced in material because of each of these three vibrations. 4

(c) A machine mounted on springs and fitted with a dashpot has a mass of 60 kg. There are three springs each of stiffness 12 N/mm. The amplitude of vibration reduces from 45 mm to 8 mm in two complete oscillations. Assuming that the damping forces varies as the velocity, determine the—

(i) damping coefficient;

(ii) ratio of frequencies of damped and undamped vibrations;

(iii) periodic time of damped vibration. 10

(d) A vertical steel shaft of 15 mm diameter is held in long bearing 1 m apart and carries at its middle a disc of mass 15 kg. The eccentricity of the centre of gravity of the disc from the centre of the shaft is 0·30 mm. The modulus of elasticity of the shaft material is 200 GPa and the permissible stress is 70 MPa. Determine the critical speed of the shaft neglecting the mass of the shaft and the range of speed over which it is unsafe to run the shaft. [Assume fixed end beam] 10

(Turn Over)

5. (a) What are turning moment diagrams? 2

(b) Derive an expression for the inertia force acting on reciprocating engine due to acceleration or retardation of machine elements. 4

(c) The torque exerted on the crankshaft of a two-stroke engine is given by

$$T = (15000 + 2000 \sin 2\theta - 1800 \cos 2\theta) \text{ N-m}$$

Assuming the resistance torque to be constant, determine—

(i) the power of the engine when running at 150 r.p.m.;

(ii) the moment of inertia of flywheel if the speed variation from the mean speed of 150 r.p.m. is not to exceed $\pm$ 0·5%;

(iii) the angular acceleration of flywheel for $\theta = 30°$. 10

(d) A horizontal steam engine running at 240 r.p.m. has a bore of 20 cm and stroke of 36 cm. The piston rod is 2 cm in diameter and connecting rod length is 90 cm. The mass of reciprocating parts is 7 kg and the frictional resistance is equivalent to a force of 500 N. Determine the following when the crank is at 120° from the inner dead centre, the mean pressure being

50×10^3 N/m^2 on the cover side and 1×10^3 N/m^2 on crank side 10

(i) Thrust on the connecting rod

(ii) Thrust on the cylinder walls

(iii) Load on the bearings

(iv) Turning moment on the crankshaft